

CKI: a Ka/Ks-inspired index

Quantifying functional cell-type divergence, normalized to the housekeeping baseline

compare expression profiles

1

Cell state A

Cell state B

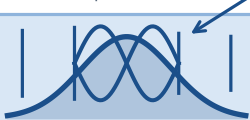
2

Pseudobulk expression profiles



3

Housekeeping genes
Neutral expression baseline



k_n (JS divergence)

Functional genes
Under selective pressure



k_f (JS divergence)

4

$$\omega = k_f/k_n$$

$\omega > 1 \rightarrow$ divergence exceeds neutral baseline

5



Permutation test

Strong

Moderate

Weak

Applications

TCGA cancer

Tumor vs. normal

Tumor Normal

Human brain atlas

Cell type diversity

Tabula atlases

Validation across tissues

Key insight:

ω distinguishes functional remodeling from neutral transcriptomic drift